Lingfan Bao

PhD Candidate in Computer Science, UCL · Humanoid Whole-Body Control & Motion Generation

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Summary

- **Research:** PhD candidate in Computer Science at UCL, working on **legged robots** (quadrupeds, bipeds and humanoids) since 2021, moving from model-based control (trajectory optimisation, MPC, ILC) to learning-based methods (reinforcement learning, imitation learning, diffusion models).
- **Hardware:** extensive hands-on experience on real robots, from classical methods (trajectory optimisation, MPC and ILC for jumping on Unitree A1) to current learning-based ones (RL whole-body control on Unitree G1); reproduced and extended the HECTOR biped.
- **Publications:** several first / co-first author papers in leading venues including Artificial Intelligence Review, ICRA and CDC, plus a book chapter to appear; co-authored papers at RSS and Humanoids.
- **Beyond research:** ATEC finalist (one of two humanoid teams among 12); administrator of Physical I/O, a London Physical AI community; UCL Robotics Society Research Mentor; UCL–NVIDIA Robotics Day co-organiser.

Education

University College London (UCL) Ph.D. in Computer Science, expected Aug 2027. *Nov 2023 ∼ Present*

· **Research areas:** humanoid whole-body control and motion generation — how human motion can be generated, retargeted and executed on real humanoid robots while keeping its expressive character.

· **Advisor:** Prof. Chengxu Zhou

University of Southern California (USC) M.S. in Mechanical Engineering. GPA 3.883/4.00. *Jan 2021 ∼ Jan 2023*

· **Research:** agile jumping for legged robots (trajectory optimisation, MPC, ILC)

· **Advisor:** Prof. Quan Nguyen

TU Dublin B.Eng. in Mechanical Engineering, First-Class Honours. GPA 3.71/4.00. *Sep 2019 ∼ Jun 2020*

Nanjing Tech University B.Eng. in Mechanical Engineering. GPA 3.50/4.00. *Sep 2016 ∼ Jun 2019*

Research Experience

Ph.D. Research @ UCL Humanoid Robotics Lab Advisor: Prof. Chengxu Zhou *Nov 2023 ∼ Present*

· Beyond Kinematic Tracking: gated residual policies on frozen whole-body trackers (PPO teachers + DAgger) that keep the amplitude, energy and posture of human motion on a real humanoid. Paper #7

· HIAER: a hierarchy of VLM intent inference, diffusion motion generation and RL whole-body control that lets a humanoid respond to social context with expressive motion. Paper #5

· PAMoR: real-time, affect-controllable motion generation with diffusion models, letting a humanoid produce whole-body motion for a chosen emotion. Paper #6

· Motion retargeting review: introduced Retargeting Readiness Levels (RRL) to map explicit and implicit cross-embodiment retargeting by preserved fidelity and embodiment breadth, and the gap they share (force-aware translation). Paper #8

· Bipedal RL: first-authored a survey of deep RL for bipedal locomotion (AI Review) and a book chapter on sim-to-real transfer; co-authored bipedal walking on a quadruped via adversarial motion priors (TAROS 2024) and gait-conditioned RL (Humanoids 2025). Papers #3–4, #9–10

· Planning and platforms: long-horizon walking optimisation under dynamic, friction and kinematic constraints; reproduced and extended the HECTOR bipedal platform.

M.S. Research @ USC Dynamic Robotics and Control Lab Advisor: Prof. Quan Nguyen *Jun 2021 ∼ Jun 2023*

· Stepping-stone jumping: full-body trajectory optimisation (TO) plans the jumps offline, and single-rigid-body model predictive control (MPC) links them online as real-time feedback; continuous stepping-stone jumping in simulation and single jumps over unknown ground height on hardware. Paper #1

· ILC agile jumping: multi-stage iterative learning control (ILC) that updates feedforward contact forces from previous-trial trajectory errors, learning simple jumps first and transferring the skill to harder targets; validated in simulation and on Unitree A1 hardware. Paper #2

Research Assistant @ IVIP Lab, Tsinghua University Advisor: Prof. Fei Qiao *Sep 2018 ~ May 2019*

· Low-cost 3D-printed quadruped: designed the hardware, circuits and embedded integration; validated walking gaits and tested obstacle avoidance.

· Other projects: handwritten-digit recognition with a dynamic vision sensor (DVS); mechanism design and Arduino / Raspberry Pi integration for a rescue robot.

Skills

Learning & control: PPO, DAgger, residual policies, whole-body control, imitation learning, sim-to-real

Optimisation: trajectory optimisation, MPC, ILC, contact scheduling

Generation: diffusion models, motion retargeting

Simulation & robots: MuJoCo, Isaac Sim / Isaac Lab, ROS; hardware deployment on Unitree G1, A1 and HECTOR

Programming: Python, C++, MATLAB; PyTorch; Linux, Git; SolidWorks, Creo / Pro.E

AI coding tools: Claude Code and Codex for prototyping, experiment scripts and refactoring

Languages: English (working language), Mandarin Chinese (native)

Selected Publications

Asterisks (*) denote equal contribution. Each group in chronological order. Full list on Google Scholar.

MAIN PAPERS

1. **Continuous Jumping for Legged Robots on Stepping Stones via Trajectory Optimization and Model Predictive Control**
   C. Nguyen*, **L. Bao***, Q. Nguyen — *IEEE CDC 2022.*
2. **Mastering Agile Jumping Skills from Simple Practices with Iterative Learning Control**
   C. Nguyen*, **L. Bao***, Q. Nguyen — *IEEE ICRA 2025.*
3. **Deep Reinforcement Learning for Robotic Bipedal Locomotion: A Brief Survey**
   L. Bao, J. Humphreys, T. Peng, C. Zhou — *Artificial Intelligence Review, 59: 38, 2026.*
4. **Sim-to-Real Transfer in Deep Reinforcement Learning for Bipedal Locomotion**
   L. Bao, T. Peng, C. Zhou — *Book chapter, to appear.*
5. **Hierarchical Intention-Aware Expressive Motion Generation for Humanoid Robots**
   L. Bao, Y. Pan, T. Peng, D. Kanoulas, C. Zhou — *Int. J. Social Robotics, under review.*
6. **PAMoR: Parameterized Affective Motion Generation in Real Time for Humanoid Robots**
   Y. Pan, **L. Bao**, T. Peng, C. Zhou — *IEEE RA-L, under review.*
7. **Beyond Kinematic Tracking: Preserving Expressive Characteristics in Humanoid Whole-Body Control**
   L. Bao, H. Zhu, Y. Pan, J. Wang, J. Yu, T. Peng, Y. Jia, D. Kanoulas, C. Zhou — *IEEE RA-L, under review.*
8. **Motion Retargeting for Physical AI: The Data Translation Layer from Biological Motion to Robot Skill**
   L. Bao, T. Peng, Y. Jia, C. Zhou — *Review, in preparation.*

CO-AUTHORED PAPERS

9. **Learning Bipedal Walking on a Quadruped Robot via Adversarial Motion Priors**
   T. Peng, **L. Bao**, J. Humphreys, A. M. Delfaki, D. Kanoulas, C. Zhou — *TAROS 2024.*
10. **Gait-Conditioned Reinforcement Learning with Multi-Phase Curriculum for Humanoid Locomotion**
    T. Peng, **L. Bao**, C. Zhou — *IEEE-RAS Humanoids 2025.*
11. **SanD-Planner: Sample-Efficient Diffusion Planner in B-Spline Space for Robust Local Navigation**
    J. Wang, **L. Bao**, T. Yang, D. Martinez Plasencia, J. Jiao, D. Kanoulas — *RSS 2026.*

Other Experience

Teaching: Teaching Assistant at UCL East for Legged Robotics System and Mechatronics (Oct 2024 ~ Present).

Talks: UCL–NVIDIA Robotics Day (co-organiser and speaker, Feb 2026); UCL Robotics Society (Mar 2026).

Peer reviewing: ICRA, IROS, IEEE T-RO, IEEE RA-L and IEEE/ASME TMECH.

Community: administrator of Physical I/O (a Physical AI community); UCL Robotics Society Research Mentor.

ATEC: one of two humanoid teams among 12 finalists in outdoor autonomous locomotion and manipulation.